SYSEN 5888 Spring 2026

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Homework 2, Question 2

Goal: ConvNets while renowned for their prowess in image processing, have also demonstrated strong capabilities in handling sequential data such as text. In this problem, you will be applying these principles of CNNs to a classic problem in natural language processing - sentiment analysis.

Tools: Numpy, PyTorch, Keras (TensorFlow)

Data: IMDB movie reviews dataset provided by `tensorflow.keras.datasets.imdb`

Task: Load the IMDB dataset from Keras using a vocabulary size of 2000 for tokenization & numericalization. Each review in the dataset is already pre-processed and encoded as a sequence of word indexes. A mapping between words and their corresponding indexes is provided using the `imdb.get_word_index()` method. For consistent input to the model, your task is to pad the reviews or truncate them to a uniform length. This can be achieved using the `pad_sequences` method from Keras to convert all reviews to a length of 300 words using the `maxlen` argument in the `pad_sequences` method. The preprocessed NumPy arrays will then be fed into a PyTorch CNN for sentiment classification.

In [2]:
```
# Update packages in Colab server to latest stable versions
%pip install --upgrade torch tensorflow numpy pandas matplotlib
```

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Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.1
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Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-pac
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(from markdown-it-py>=2.2.0->rich->keras>=3.10.0->tensorflow) (0.1.2)
```

In [3]:

```python
# Import necessary libraries
import os
import numpy as np
import matplotlib.pyplot as plt
import torch
import torch.nn as nn
import torch.optim as optim
import tensorflow as tf
import pandas as pd
from torch.utils.data import DataLoader, TensorDataset
from tensorflow.keras.datasets import imdb
from tensorflow.keras.preprocessing.sequence import pad_sequences

# Hyperparameters for data processing
VOCAB_SIZE = 2000  # per homework
MAX_LEN = 300      # fixed sequence length
BATCH_SIZE = 32    # per homework
SHUFFLE_SEED = 42

# Load IMDB dataset from Keras (already tokenized and indexed)
(X_train_full, y_train_full), (X_test, y_test) = imdb.load_data(num_words=VOCAB_SIZ

# Get the word -> index mapping (for analysis / interpretability)
word_index = imdb.get_word_index()
```

```
# Pad and truncate to fixed length of 300 tokens
X_train_full = pad_sequences(X_train_full, maxlen=MAX_LEN, padding="post", truncati
X_test = pad_sequences(X_test, maxlen=MAX_LEN, padding="post", truncating="post")

# Create validation split from training data
np.random.seed(SHUFFLE_SEED)
indices = np.random.permutation(len(X_train_full))
X_train_full = X_train_full[indices]
y_train_full = np.array(y_train_full)[indices]

n_val = 1000 # per homework
X_val = X_train_full[:n_val]
y_val = y_train_full[:n_val]
X_train = X_train_full[n_val:]
y_train = y_train_full[n_val:]

# Summary
print(f"Training samples: {len(X_train)}, Validation samples: {len(X_val)}, Test sa
print(f"Sequence length: {MAX_LEN}, Batch size: {BATCH_SIZE}, Vocab size (num_words
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/im
db.npz
17464789/17464789 ━━━━━━━━━━━━━━━━━━━━ 1s 0us/step
/usr/local/lib/python3.12/dist-packages/numpy/lib/_format_impl.py:838: VisibleDeprec
ationWarning: dtype(): align should be passed as Python or NumPy boolean but got `al
ign=0`. Did you mean to pass a tuple to create a subarray type? (Deprecated NumPy 2.
4)
  array = pickle.load(fp, **pickle_kwargs)
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/im
db_word_index.json
1641221/1641221 ━━━━━━━━━━━━━━━━━━━━ 1s 1us/step
Training samples: 24000, Validation samples: 1000, Test samples: 25000
Sequence length: 300, Batch size: 32, Vocab size (num_words): 2000
```

Architecture:

The architecture of the convolutional neural network model for this problem is as follows:

Embedding Layer: Input Vocabulary Size: 2000 words Embedding Dimension: 16 Input Length: 300 words

Conv1D Layer: Filters: 128 Kernel Size: 3 Activation: ReLU Stride: 1 Padding: Valid

GlobalMaxPooling1D Layer

Dense Layer: Units: 1 Activation: Sigmoid

In [4]:
```
# PyTorch CNN Model Definition
class TextCNN(nn.Module):
    def __init__(
        self,
        vocab_size: int,
        embed_dim: int = 16,
        num_filters: int = 128,
        kernel_size: int = 3,
```

```
    ) -> None:
        super().__init__()
        # Embedding layer: maps word indices to dense vectors
        self.embedding = nn.Embedding(vocab_size, embed_dim, padding_idx=0)

        # 1D convolution over the sequence (time) dimension
        self.conv = nn.Conv1d(
            in_channels=embed_dim,
            out_channels=num_filters,
            kernel_size=kernel_size,
        )

        # Global max pooling over the time dimension
        self.global_max_pool = nn.AdaptiveMaxPool1d(output_size=1)

        # Final classification layer
        self.fc = nn.Linear(num_filters, 1)

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        """Forward pass.

        x: LongTensor of shape (batch_size, seq_len)
        returns: probabilities of shape (batch_size,)
        """
        # (batch, seq_len) -> (batch, seq_len, embed_dim)
        embedded = self.embedding(x)
        # (batch, seq_len, embed_dim) -> (batch, embed_dim, seq_len)
        embedded = embedded.permute(0, 2, 1)

        # Convolution + ReLU
        conv_out = torch.relu(self.conv(embedded))  # (batch, num_filters, L')

        # Global max pooling over time -> (batch, num_filters, 1)
        pooled = self.global_max_pool(conv_out).squeeze(-1)  # (batch, num_filters)

        # Linear layer to a single logit
        logits = self.fc(pooled).squeeze(-1)  # (batch,)

        # Sigmoid for binary sentiment probability
        probs = torch.sigmoid(logits)
        return probs


# Instantiate model (will be moved to appropriate device in training cell)
EMBED_DIM = 16
NUM_FILTERS = 128
KERNEL_SIZE = 3

# VOCAB_SIZE is defined in the preprocessing cell
model = TextCNN(vocab_size=VOCAB_SIZE, embed_dim=EMBED_DIM, num_filters=NUM_FILTERS
print(model)
```

```
TextCNN(
  (embedding): Embedding(2000, 16, padding_idx=0)
  (conv): Conv1d(16, 128, kernel_size=(3,), stride=(1,))
  (global_max_pool): AdaptiveMaxPool1d(output_size=1)
  (fc): Linear(in_features=128, out_features=1, bias=True)
)
```

Training: The model should be compiled using the 'binary_crossentropy' as the loss function and 'adam' optimizer. Additionally, 'accuracy' should be assigned as the main metric. A subset of the training data (1000 samples) should be set aside as a validation set, while the rest should be used for training. The model should be trained for a total of 30 (or 10) epochs, with a batch size of 32. After training, the model should be evaluated on the test data to obtain the final accuracy score. This will give a measure of how well the model can generalize to unseen reviews.

In [5]:
```python
## Helper Functions

# BASIC TRAIN / EPOCH
# Mirroring HW2_Q1 structure but adapted for binary sentiment labels.
def train(model, loader, criterion, optimizer, device):
    """Train the model for one epoch."""
    model.train()
    running_loss = 0.0
    correct = 0
    total = 0

    for inputs, labels in loader:
        inputs = inputs.to(device)
        # BCE loss expects floating labels in {0, 1}
        labels = labels.float().to(device)

        optimizer.zero_grad()
        outputs = model(inputs)  # (batch,)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()

        running_loss += loss.item() * inputs.size(0)

        # Convert probabilities to class predictions using 0.5 threshold
        preds = (outputs >= 0.5).float()
        total += labels.size(0)
        correct += (preds == labels).sum().item()

    epoch_loss = running_loss / len(loader.dataset)
    epoch_acc = correct / total
    return epoch_loss, epoch_acc


# BASIC EVAL / EPOCH
def evaluate(model, loader, criterion, device):
    """Evaluate the model on validation or test data."""
    model.eval()
    running_loss = 0.0
```

```
    correct = 0
    total = 0

    with torch.no_grad():
        for inputs, labels in loader:
            inputs = inputs.to(device)
            labels = labels.float().to(device)

            outputs = model(inputs)
            loss = criterion(outputs, labels)

            running_loss += loss.item() * inputs.size(0)

            preds = (outputs >= 0.5).float()
            total += labels.size(0)
            correct += (preds == labels).sum().item()

    epoch_loss = running_loss / len(loader.dataset)
    epoch_acc = correct / total
    return epoch_loss, epoch_acc


# PLOT FUNCTIONS (SAVE TO DISK LIKE HW2_Q1)

def plot_loss(curve, dataset, output_dir="Plot JPGs"):
    """Plot and save a single loss curve (training or validation)."""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    plt.figure()
    plt.plot(curve)
    plt.xlabel("Epoch")
    plt.ylabel(f"{dataset} Loss")
    plt.title(f"Binary Cross-Entropy Loss - {dataset}")
    filename = os.path.join(output_dir, f"LossCurve_{dataset}.jpg")
    plt.savefig(filename)
    plt.close()
```

In [6]:

```
# MODEL TRAIN - ALL EPOCHS
# Follows HW2_Q1-style training loop.
def train_model(model, train_loader, val_loader, criterion, optimizer, device, num_
    """Train the model and evaluate on validation data after each epoch."""
    training_loss_curve = []
    validation_loss_curve = []
    training_accuracy = []
    validation_accuracy = []

    for epoch in range(num_epochs):
        train_loss, train_acc = train(model, train_loader, criterion, optimizer, de
        val_loss, val_acc = evaluate(model, val_loader, criterion, device)

        training_loss_curve.append(train_loss)
        training_accuracy.append(train_acc)
        validation_loss_curve.append(val_loss)
        validation_accuracy.append(val_acc)
```

```
        print(
            f"Epoch {epoch + 1}/{num_epochs} | "
            f"Train Loss: {train_loss:.4f} | Train Acc: {train_acc:.4f} | "
            f"Val Loss: {val_loss:.4f} | Val Acc: {val_acc:.4f}"
        )

    return training_loss_curve, validation_loss_curve, training_accuracy, validatio
```

In [7]:

```
# MAIN EXPERIMENT HELPER
# Uses HW2_Q1-style run_experiment pattern adapted to IMDB sentiment.
def run_experiment(max_epochs=30, batch_size=BATCH_SIZE):
    """Run full experiment: build loaders, train model, evaluate on test set."""
    # Detect device (GPU on Colab if available)
    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    print(f"Using device: {device}")

    # Build PyTorch datasets and loaders from preprocessed NumPy arrays
    train_dataset = TensorDataset(
        torch.from_numpy(X_train).long(),
        torch.from_numpy(y_train).float(),
    )
    val_dataset = TensorDataset(
        torch.from_numpy(X_val).long(),
        torch.from_numpy(y_val).float(),
    )
    test_dataset = TensorDataset(
        torch.from_numpy(X_test).long(),
        torch.from_numpy(y_test).float(),
    )

    train_loader = DataLoader(train_dataset, batch_size=batch_size, shuffle=True)
    val_loader = DataLoader(val_dataset, batch_size=batch_size, shuffle=False)
    test_loader = DataLoader(test_dataset, batch_size=batch_size, shuffle=False)

    # Instantiate model fresh for this run and move to device
    model = TextCNN(
        vocab_size=VOCAB_SIZE,
        embed_dim=EMBED_DIM,
        num_filters=NUM_FILTERS,
        kernel_size=KERNEL_SIZE,
    )
    model.to(device)

    # Binary cross-entropy loss and Adam optimizer
    criterion = nn.BCELoss()
    optimizer = optim.Adam(model.parameters())

    # Train
    training_loss_curve, validation_loss_curve, training_accuracy, validation_accur
        model,
        train_loader,
        val_loader,
        criterion,
        optimizer,
        device,
        num_epochs=max_epochs,
```

```
    )

    # Final test evaluation
    test_loss, test_acc = evaluate(model, test_loader, criterion, device)

    # Save individual curves for later quad plotting
    plot_loss(training_loss_curve, "Training")
    plot_loss(validation_loss_curve, "Validation")
    plot_accuracy(training_accuracy, "Training")
    plot_accuracy(validation_accuracy, "Validation")

    # Prepare summary row
    row = {
        "Model Name": "TextCNN",
        "Epochs": max_epochs,
        "Training Accuracy (%)": training_accuracy[-1] * 100.0,
        "Validation Accuracy (%)": validation_accuracy[-1] * 100.0,
        "Test Accuracy (%)": test_acc * 100.0,
        "Final Test Loss": test_loss,
    }

    return row, {
        "train_loss": training_loss_curve,
        "val_loss": validation_loss_curve,
        "train_acc": training_accuracy,
        "val_acc": validation_accuracy,
    }
```

In [8]:

```
## Train and Run Model
# binary_crossentropy loss, ADAM optimizer
# Run for 30 epochs, batch size 32
# Evaluate on test data to obtain accuracy score

# Define results table, maximum epochs to run
results_dataframe = pd.DataFrame(
    columns=[
        "Model Name",
        "Epochs",
        "Training Accuracy (%)",
        "Validation Accuracy (%)",
        "Test Accuracy (%)",
        "Final Test Loss",
    ]
)
MAX_EPOCHS = 30

# Run experiment
row, history = run_experiment(max_epochs=MAX_EPOCHS)
results_dataframe = pd.concat([results_dataframe, pd.DataFrame([row])], ignore_inde
print(results_dataframe)
```

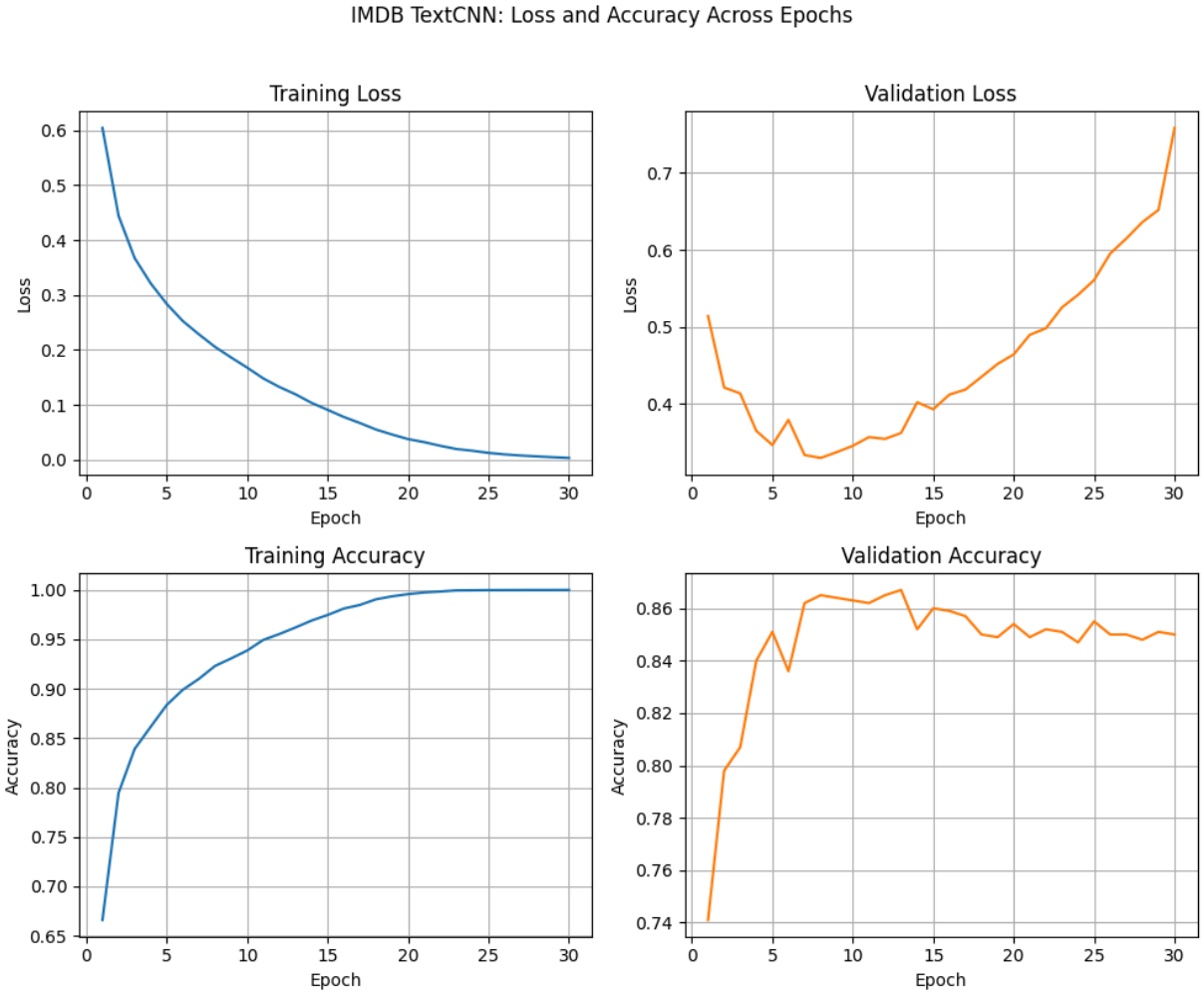
```
Using device: cuda
Epoch 1/30 | Train Loss: 0.6044 | Train Acc: 0.6659 | Val Loss: 0.5141 | Val Acc: 0.
7410
Epoch 2/30 | Train Loss: 0.4444 | Train Acc: 0.7945 | Val Loss: 0.4215 | Val Acc: 0.
7980
Epoch 3/30 | Train Loss: 0.3675 | Train Acc: 0.8390 | Val Loss: 0.4137 | Val Acc: 0.
8070
Epoch 4/30 | Train Loss: 0.3213 | Train Acc: 0.8615 | Val Loss: 0.3651 | Val Acc: 0.
8400
Epoch 5/30 | Train Loss: 0.2838 | Train Acc: 0.8837 | Val Loss: 0.3468 | Val Acc: 0.
8510
Epoch 6/30 | Train Loss: 0.2526 | Train Acc: 0.8990 | Val Loss: 0.3794 | Val Acc: 0.
8360
Epoch 7/30 | Train Loss: 0.2284 | Train Acc: 0.9101 | Val Loss: 0.3338 | Val Acc: 0.
8620
Epoch 8/30 | Train Loss: 0.2056 | Train Acc: 0.9231 | Val Loss: 0.3300 | Val Acc: 0.
8650
Epoch 9/30 | Train Loss: 0.1863 | Train Acc: 0.9307 | Val Loss: 0.3375 | Val Acc: 0.
8640
Epoch 10/30 | Train Loss: 0.1678 | Train Acc: 0.9387 | Val Loss: 0.3457 | Val Acc:
0.8630
Epoch 11/30 | Train Loss: 0.1482 | Train Acc: 0.9495 | Val Loss: 0.3570 | Val Acc:
0.8620
Epoch 12/30 | Train Loss: 0.1326 | Train Acc: 0.9554 | Val Loss: 0.3547 | Val Acc:
0.8650
Epoch 13/30 | Train Loss: 0.1193 | Train Acc: 0.9620 | Val Loss: 0.3624 | Val Acc:
0.8670
Epoch 14/30 | Train Loss: 0.1037 | Train Acc: 0.9690 | Val Loss: 0.4023 | Val Acc:
0.8520
Epoch 15/30 | Train Loss: 0.0910 | Train Acc: 0.9747 | Val Loss: 0.3932 | Val Acc:
0.8600
Epoch 16/30 | Train Loss: 0.0781 | Train Acc: 0.9812 | Val Loss: 0.4122 | Val Acc:
0.8590
Epoch 17/30 | Train Loss: 0.0671 | Train Acc: 0.9848 | Val Loss: 0.4188 | Val Acc:
0.8570
Epoch 18/30 | Train Loss: 0.0552 | Train Acc: 0.9905 | Val Loss: 0.4353 | Val Acc:
0.8500
Epoch 19/30 | Train Loss: 0.0461 | Train Acc: 0.9935 | Val Loss: 0.4521 | Val Acc:
0.8490
Epoch 20/30 | Train Loss: 0.0378 | Train Acc: 0.9958 | Val Loss: 0.4644 | Val Acc:
0.8540
Epoch 21/30 | Train Loss: 0.0322 | Train Acc: 0.9975 | Val Loss: 0.4898 | Val Acc:
0.8490
Epoch 22/30 | Train Loss: 0.0256 | Train Acc: 0.9984 | Val Loss: 0.4983 | Val Acc:
0.8520
Epoch 23/30 | Train Loss: 0.0197 | Train Acc: 0.9996 | Val Loss: 0.5255 | Val Acc:
0.8510
Epoch 24/30 | Train Loss: 0.0165 | Train Acc: 0.9997 | Val Loss: 0.5418 | Val Acc:
0.8470
Epoch 25/30 | Train Loss: 0.0126 | Train Acc: 0.9998 | Val Loss: 0.5610 | Val Acc:
0.8550
Epoch 26/30 | Train Loss: 0.0099 | Train Acc: 0.9999 | Val Loss: 0.5955 | Val Acc:
0.8500
Epoch 27/30 | Train Loss: 0.0079 | Train Acc: 0.9999 | Val Loss: 0.6149 | Val Acc:
0.8500
Epoch 28/30 | Train Loss: 0.0063 | Train Acc: 1.0000 | Val Loss: 0.6362 | Val Acc:
```

```
0.8480
Epoch 29/30 | Train Loss: 0.0048 | Train Acc: 1.0000 | Val Loss: 0.6521 | Val Acc:
0.8510
Epoch 30/30 | Train Loss: 0.0035 | Train Acc: 1.0000 | Val Loss: 0.7586 | Val Acc:
0.8500
  Model Name Epochs Training Accuracy (%) Validation Accuracy (%)  \
0    TextCNN     30                 100.0                    85.0

  Test Accuracy (%) Final Test Loss
0            84.904        0.742207
```

Visualization: Plot the accuracy and loss for both training and validation datasets across epochs to analyze the performance of the model over epochs.

In [9]:

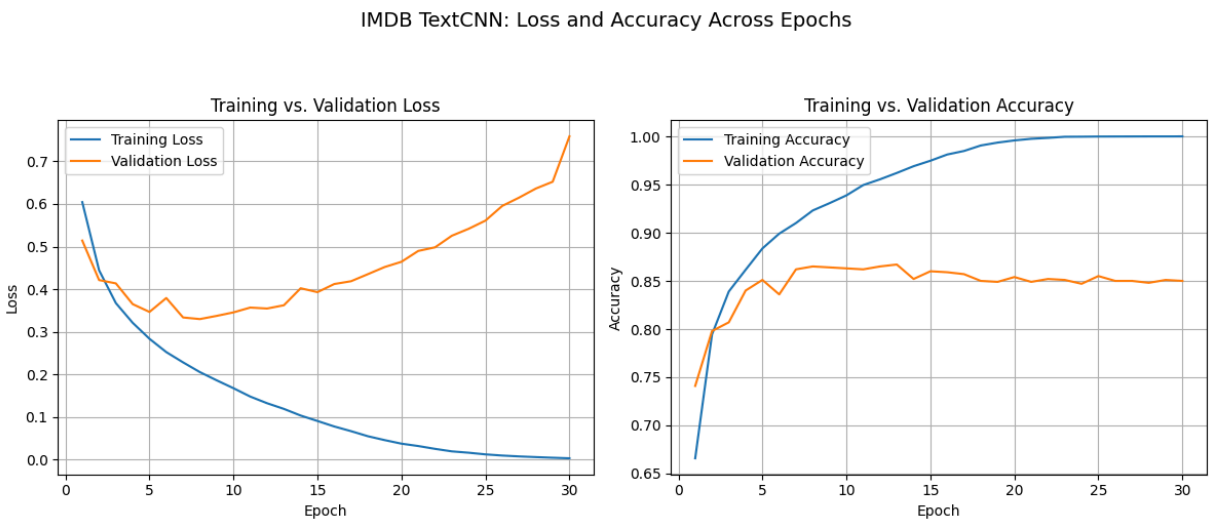
```python
# Plots across epochs
# Plot accuracy and loss for both training and validation - build a quad
# Uses history dict returned by run_experiment()

epochs = range(1, len(history["train_loss"]) + 1)

fig, axes = plt.subplots(2, 2, figsize=(10, 8))

# Top-left: Training Loss
axes[0, 0].plot(epochs, history["train_loss"], color="C0")
axes[0, 0].set_xlabel("Epoch")
axes[0, 0].set_ylabel("Loss")
axes[0, 0].set_title("Training Loss")
axes[0, 0].grid(True)

# Top-right: Validation Loss
axes[0, 1].plot(epochs, history["val_loss"], color="C1")
axes[0, 1].set_xlabel("Epoch")
axes[0, 1].set_ylabel("Loss")
axes[0, 1].set_title("Validation Loss")
axes[0, 1].grid(True)

# Bottom-left: Training Accuracy
axes[1, 0].plot(epochs, history["train_acc"], color="C0")
axes[1, 0].set_xlabel("Epoch")
axes[1, 0].set_ylabel("Accuracy")
axes[1, 0].set_title("Training Accuracy")
axes[1, 0].grid(True)

# Bottom-right: Validation Accuracy
axes[1, 1].plot(epochs, history["val_acc"], color="C1")
axes[1, 1].set_xlabel("Epoch")
axes[1, 1].set_ylabel("Accuracy")
axes[1, 1].set_title("Validation Accuracy")
axes[1, 1].grid(True)

plt.suptitle("IMDB TextCNN: Loss and Accuracy Across Epochs", fontsize=12, y=1.02)
plt.tight_layout()
plt.show()
```



In [10]:

```python
# Plot loss and accuracy on top of each other
epochs = range(1, len(history["train_loss"]) + 1)

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Left subplot: Training and Validation Loss
axes[0].plot(epochs, history["train_loss"], label="Training Loss", color="C0")
axes[0].plot(epochs, history["val_loss"], label="Validation Loss", color="C1")
axes[0].set_xlabel("Epoch")
axes[0].set_ylabel("Loss")
axes[0].set_title("Training vs. Validation Loss")
axes[0].legend()
axes[0].grid(True)

# Right subplot: Training and Validation Accuracy
axes[1].plot(epochs, history["train_acc"], label="Training Accuracy", color="C0")
axes[1].plot(epochs, history["val_acc"], label="Validation Accuracy", color="C1")
axes[1].set_xlabel("Epoch")
axes[1].set_ylabel("Accuracy")
axes[1].set_title("Training vs. Validation Accuracy")
axes[1].legend()
axes[1].grid(True)

plt.suptitle("IMDB TextCNN: Loss and Accuracy Across Epochs", fontsize=14, y=1.05)
```

```
plt.tight_layout(rect=[0, 0.03, 1, 0.98]) # Adjust layout to prevent suptitle overl
plt.show()
```



Deliverables:

1. Model Accuracy and Loss Curves: A detailed report of the performance of the model, focusing on accuracy and loss curves.
2. Analysis of Model Performance: A thorough analysis should be conducted to discuss the results obtained from the model. This analysis should include

2a. Whether the model overfits or underfits the training data.

2b. Examination of the loss and accuracy curves to identify potential indicators of the model's behavior (such as plateaus or sharp changes).

Performance Discussion

The loss curves show training loss decreasing smoothly to 0 over 30 epochs, the validation curve in a U-shape, the training accuracy increasing smoothly to plateau at 1, and validation accuracy hovering around 85%. Test accuracy as reported textually was just shy of that 85%.

These curves clearly show the model overfitting. The validation loss climbs after Epoch 7 and does not stop. This divergence is a key hallmark of overfit models. Methods to improve the model fit include adding an early stopping metric and increasing dropout.